

Clario — Client Management Made Simple

Clario is a web-based platform designed for small service-based businesses to manage their clients and daily operations in one place. The platform combines client records, appointment scheduling, visit documentation, forms, product purchases, and document management, while also providing clients with a personal area to manage appointments, access relevant information, and prepare documents for reimbursement when applicable.

2. Product Specification

1. What Problem Does the Product Solve?

Small service-based businesses often manage their daily operations using several disconnected tools. Appointments may be managed through calendars, client communication through WhatsApp, and client information, treatment history, forms, purchases, and documents through spreadsheets, notebooks, or separate files.

This makes client management inefficient, increases administrative work, and makes it difficult to maintain a complete and organized history for each client. Clients also depend on the business for simple actions such as scheduling appointments, accessing documents, or finding information from previous visits.

Clario solves this problem by providing one centralized web-based platform for managing the ongoing relationship between a service-based business and its clients. Businesses can manage clients, appointments, visits, forms, products, and documents in one place, while clients receive a personal area where they can independently manage appointments, access relevant information, repurchase products, and prepare documents for reimbursement when applicable.

2. Who Are the Product Users?

Clario has two main types of users:

Business Owner / Service Provider - The primary users are owners of small service-based businesses that maintain ongoing relationships with their clients, such as beauticians, therapists, nutritionists, acupuncturists, private healthcare providers, and other independent service providers. Business users can manage their clients and appointments, maintain client records, document visits and treatments, add recommendations, manage products, send forms, and upload relevant documents.

End Client - Clients receive a personal area that allows them to interact with the business independently. They can book and manage appointments, complete forms, access information and documents shared by the service provider, view previously recommended or purchased products, repurchase products, and prepare relevant documents for reimbursement requests when applicable.

3. Who Is the Customer?

The paying customer is the business owner or service provider. Clario targets small service-based businesses that need to maintain an ongoing relationship and history with their clients, including beauty clinics, therapists, nutritionists, acupuncturists, private healthcare providers, personal trainers, and similar independent businesses.

Clario can operate as a Software as a Service (SaaS) product, where each business receives its own workspace and pays a recurring subscription fee. End clients use the platform as part of the service provided by the business but are not the primary paying customers.

4. What Are the Business Goals of the Product?

- **Reduce administrative work** – Reduce the time business owners spend managing appointments, searching for information, maintaining records, forms, and documents.
- **Centralize client management** – Keep each client's information, history, appointments, documents, recommendations, and purchases organized in one place.
- **Improve the client experience** – Allow clients to independently perform common actions such as booking appointments, completing forms, accessing documents, and repurchasing products.
- **Simplify reimbursement preparation** – Help clients organize the documents required for reimbursement requests following eligible private services.
- **Increase revenue opportunities** – Support product sales and make repeat purchases easier for existing clients.
- **Improve client retention** – Enable more personalized and continuous service by maintaining a complete client history.
- **Create a scalable SaaS product** – Build a platform that can serve multiple independent businesses through a subscription-based model.

5. What Software Capabilities Need to Be Built to Support the Business Goals?

User Management and Authentication- The system will support registration and login for business owners and clients, with different permissions according to the user's role.

Client Management and Client Records - Business owners will be able to create, search, and manage clients. Each client will have a dedicated record containing personal information, appointment and visit history, professional notes, recommendations, forms, purchases, and relevant documents.

Appointment and Calendar Management - Business owners will be able to define availability and manage appointments through an internal calendar. Clients will be able to view available time slots, book and manage appointments, and add confirmed appointments to a personal calendar such as Google Calendar.

Visit and Treatment Documentation - After a visit, the service provider will be able to create a summary containing relevant information such as the service provided, professional notes, recommendations, follow-up information, and related documents.

Forms and Declarations- Businesses will be able to assign forms such as health declarations, consent forms, or questionnaires to clients. Clients will complete them through their personal area, and the submitted information will become part of their client record.

Products and Purchases- Businesses will be able to manage products and associate recommendations or purchases with clients. Clients will be able to view previously recommended or purchased products and repurchase available products.

Document and Reimbursement Management -Businesses will be able to associate visit summaries, receipts, and other documents with a client's visit. Clients will be able to access relevant documents and prepare an organized package for submission to an insurance provider or healthcare organization when applicable.

6. What Are the Main Processes the Product Allows Users to Perform?

Registration and Login - Business owners and clients can register and log in. The system identifies the user's role and provides access to the appropriate interface and information.

Adding and Managing Clients - A business owner can add a client and create a dedicated client record. All future appointments, visits, forms, recommendations, purchases, and documents related to that client can then be accessed from one place.

Appointment Booking - The business owner defines available appointment times, and clients can select and book an available slot. Once confirmed, the appointment appears in

both the business calendar and the client's personal area, with an option to add it to a personal calendar.

Documenting a Visit or Treatment- After an appointment, the business owner can document the visit, including the service provided, professional notes, recommendations, products, and follow-up information. Relevant documents or receipts can also be attached to the visit.

Completing Forms and Declarations- The business can assign a form to a client, who completes it through their personal area. The submitted form is then stored as part of the client's record.

Product Repurchase - Products recommended or purchased during previous visits are associated with the client's record. The client can later view these products and repurchase available items directly through Clario.

Preparing Documents for Reimbursement - For an eligible private service, the service provider uploads the relevant visit summary, receipt, and documents. The client can access the visit through their personal area, select the required documents, and download them as an organized package for submission to the relevant insurance provider or healthcare organization.

3. Software Architecture Design

1. What Components Will Be Included in the System?

Clario will consist of five main components:

- **Frontend** – A responsive web interface for business owners and clients, built with Next.js, React, and TypeScript.
- **Backend / Server Logic** – Processes user actions, applies business rules, validates requests, checks permissions, and communicates with the database.
- **Database** – A relational PostgreSQL database for storing the application's structured data.
- **Authentication and Authorization** – Identifies logged-in users and controls which information and actions are available to each user.
- **File Storage** – Securely stores uploaded files such as receipts and visit-related documents.

Next.js will handle the frontend and server-side logic, while Supabase will provide the database, authentication, authorization capabilities, and file storage.

2. Will the System Use a Database?

Yes. Clario will use a relational PostgreSQL database provided by Supabase. A relational database is suitable because Clario contains structured data with clear relationships. For example, one business can manage multiple clients, and each client can have multiple appointments, visits, purchases, forms, and documents. Uploaded files will be stored separately in Supabase Storage, while the database will store their metadata and relationships to clients and visits.

3. What Are the Main Database Tables / Entities?

The main database entities will be:

- **Profiles** – Stores application information about authenticated users, including their role. Authentication credentials are managed by Supabase Auth.
- **Businesses** – Represents each business using Clario and stores its basic information and ownership.
- **Clients** – Stores the clients managed by each business. A client may optionally be linked to a registered Clario account, allowing businesses to manage clients even before they create an account.
- **Appointments** – Stores scheduled appointments, including the business, client, date, time, and status.
- **Visits** – Stores information documented after completed appointments, including visit summaries, professional notes, recommendations, and follow-up information.
- **Business Availability** – Stores the working hours and appointment availability defined by each business.
- **Documents** – Stores metadata and relationships for client and visit-related documents, while the actual files are stored in Supabase Storage.
- **Forms and Form Submissions** – Forms represent reusable templates such as health declarations or consent forms, while submissions store responses completed by individual clients.
- **Products and Product Recommendations** – Products represent items offered by a business, while recommendations connect products to specific clients and visits.
- **Purchases and Purchase Items** – Purchases represent client orders, while purchase items represent the individual products included in each order.

The database relationships are centered around the **business and client**. A business manages its own clients and resources, while each client can have multiple appointments, visits, forms, documents, recommendations, and purchases.

4. What Pages Will Be Included in the Application?

Clario will provide separate interfaces for business owners and clients.

Public Pages

- **Landing Page** – Introduces Clario and its main value.
- **Login** – Allows existing users to sign in.
- **Sign Up** – Allows new users to create an account.

Business Owner Pages

- **Dashboard** – Overview of upcoming appointments, recent activity, and common actions.
- **Clients** – Client list with search and client creation.
- **Client Details** – The complete client record, including personal details, appointment history, visit documentation organized by date, completed forms and declarations, documents and receipts, product recommendations, and purchase history.
- **Calendar / Appointments** – Business calendar for managing appointments and availability.
- **Products** – Management of products offered by the business.
- **Forms** – Management of reusable forms and declarations, client assignments, and completion status.
- **Business Settings** – Business and account settings.

Client Pages

- **Client Dashboard** – Overview of upcoming appointments, pending forms, and recent activity.
- **My Appointments** – Upcoming and previous appointments.
- **Book Appointment** – Allows clients to view available time slots and book appointments.

- **My Visits / Documents** – Visit history, shared summaries, receipts, and documents relevant to reimbursement preparation.
- **My Forms** – Assigned and previously submitted forms.
- **Products** – Recommended and previously purchased products, including repeat purchases when available.
- **Profile** – Personal account information.

Navigation and available actions will depend on the user's role.

5. Will the Application Require Server Actions or API Routes?

Yes. Clario will primarily use Next.js Server Actions for internal operations performed by users within the application.

Server Actions will be used for operations such as:

- Creating and updating clients.
- Booking, modifying, and cancelling appointments.
- Creating visit summaries.
- Assigning and submitting forms.
- Managing products, purchases, and documents.

Before modifying data, the server will verify the authenticated user, check permissions, validate the request, and apply the relevant business rules.

API Routes / Route Handlers will be used when a dedicated HTTP endpoint is required, particularly for external integrations. For example, if online payments are implemented, an API Route may receive notifications from the payment provider.

Therefore, most core Clario operations will use Server Actions, while API Routes will be added only when required.

6. How Will Information Flow Between the Frontend, Backend, and Database?

The general data flow will be:

User → Frontend → Backend → Database / Storage → Backend → Frontend

The Frontend displays application data and collects user input. User actions are sent to the Backend, which verifies authentication and permissions, validates the data, and applies

the relevant business rule .Valid operations are then read from or written to the Supabase PostgreSQL database. Uploaded files are stored in Supabase Storage.

The result is returned to the Frontend, which displays the updated information to the user.

7. What Users and Permissions Will Exist in the System?

Clario will initially support two main user roles:

Business Owner / Service Provider

The business owner can:

- Manage clients, appointments, and availability for their business.
- Create and manage visit summaries, forms, and client documents.
- Manage products, recommendations, and purchases.
- Access information related to their own business.

A business owner cannot access data belonging to another business.

Client

The client can:

- View, book, and manage their own appointments.
- Complete assigned forms.
- View visit information and documents shared with them.
- View recommendations and purchases and repurchase available products.
- Access their own documents for reimbursement preparation.
- Manage their personal profile.

Clients cannot access another client's information or modify professional records created by the service provider.

Permissions will be enforced both at the application level and through Supabase Row Level Security (RLS), ensuring that users can only access records they are authorized to view or modify.

8. Which External Libraries or Services Will Be Integrated, and Why?

Clario will use the following core technologies and supporting services:

- **Next.js and TypeScript** – Used to build the web application and its server-side logic with a structured and type-safe codebase.
- **Supabase** – Provides the PostgreSQL database, authentication, access control, and file storage.
- **Vercel** – Used to deploy and host the application through a public URL.
- **shadcn/ui** – May be used for reusable UI components and a consistent interface.
- **Zod** – May be used for validating user input before it is processed or stored.
- **Calendar Integration / UI Library** – Clario will maintain its own appointment calendar. A calendar library may be used for the interface, and clients will be able to add confirmed appointments to a personal calendar such as Google Calendar.
- **Payment Provider (Optional)** – If online product payments are implemented, a provider such as Stripe may be integrated to securely process payments.

Next.js, TypeScript, Supabase, and Vercel will form the core technology stack, while optional libraries and integrations will be selected according to the needs of the MVP.

4. Detailed Technical Design

1. Project Folder Structure

Clario will follow a modular folder structure based on the Next.js App Router, separating application pages, reusable UI components, server-side operations, and shared infrastructure.

- **app/ – Application Pages and Routing:** Contains all application pages, organized into three main areas:
 - **(auth)** – Login and sign-up pages.
 - **(business)** – Business-owner pages, including the dashboard, clients and client records, calendar, products, forms, and settings.
 - **(client)** – Client pages, including the dashboard, appointments, booking, visits and documents, forms, products, and profile.
 - **api/** – Route Handlers required for external integrations.
- **components/ – Reusable UI Components:** Contains general UI elements and feature-specific components for clients, appointments, visits, forms, products, and documents.

- **actions/ – Server Actions:** Contains server-side operations organized by feature.
- **lib/ – Shared Infrastructure and Utilities:** Contains Supabase configuration, validation logic, and shared utility functions.
- **types/ – TypeScript Types:** Contains shared TypeScript type definitions.
- **public/ – Static Assets:** Contains images, logos, and other static files.

This separation keeps pages, interface components, server operations, and shared technical functionality organized and easy to maintain.

2. Main Component Structure

Clario's user interface will be divided into reusable React components organized by feature. Pages will combine these components, while shared components can be reused across multiple areas of the application.

- **Client Components** – ClientCard, ClientForm, ClientHeader, and components for personal details and client history.
- **Appointment Components** – AppointmentCalendar, AppointmentCard, BookingForm, and AvailabilitySettings.
- **Visit Components** – VisitTimeline and VisitSummary for documenting and displaying visits chronologically.
- **Form Components** – FormBuilder, FormList, and FormSubmission for creating, assigning, completing, and displaying forms.
- **Document Components** – DocumentUpload and DocumentList for managing client-related documents.
- **Product Components** – ProductCard, ProductForm, and RecommendationList for products, recommendations, and purchases.
- **Shared UI Components** – Reusable elements such as buttons, input fields, dialogs, cards, and navigation components.

Larger pages will combine several of these components. For example, the Client Details page will use components for personal details, visit history, forms, documents, recommendations, and purchases.

3. Database Structure - Clario will use a relational PostgreSQL database managed through Supabase. The database will be organized around businesses and their clients, with related tables for appointments, visits, forms, documents, products, and purchases.

Main Tables and Fields

Table	Key Fields
Profiles	id (PK, linked to Supabase Auth), full_name, email, role, created_at
Businesses	id (PK), owner_id (FK → Profiles), name, description, phone, email, created_at
Clients	id (PK), business_id (FK → Businesses), user_id (FK → Profiles, optional), full_name, email, phone, notes, created_at
Business Availability	id (PK), business_id (FK → Businesses), day_of_week, start_time, end_time
Appointments	id (PK), business_id (FK → Businesses), client_id (FK → Clients), start_time, end_time, status, notes, created_at
Visits	id (PK), appointment_id (FK → Appointments), client_id (FK → Clients), summary, professional_notes, follow_up, created_at
Documents	id (PK), client_id (FK → Clients), visit_id (FK → Visits, optional), type, file_path, created_at
Forms	id (PK), business_id (FK → Businesses), title, description, fields, created_at
Form Submissions	id (PK), form_id (FK → Forms), client_id (FK → Clients), answers, submitted_at
Products	id (PK), business_id (FK → Businesses), name, description, price, is_active, created_at
Product Recommendations	id (PK), visit_id (FK → Visits), client_id (FK → Clients), product_id (FK → Products), notes, created_at
Purchases	id (PK), business_id (FK → Businesses), client_id (FK → Clients), status, total_amount, created_at
Purchase Items	id (PK), purchase_id (FK → Purchases), product_id (FK → Products), quantity, unit_price

Uploaded files will be stored in Supabase Storage rather than directly in PostgreSQL. The Documents table will store the file path and associate each file with the relevant client or visit.

Main Relationships

- A **Business** belongs to a business owner and can have multiple clients, appointments, products, forms, and availability records.
- A **Client** belongs to a business and may optionally be linked to a registered Clario account. A client can have multiple appointments, visits, documents, form submissions, recommendations, and purchases.
- An **Appointment** belongs to a client and may result in a visit.
- A **Visit** may be associated with documents and product recommendations.
- A **Form** can have multiple submissions from different clients.
- A **Product** can appear in multiple recommendations and purchases.
- A **Purchase** can contain multiple products through the Purchase Items table.

4. Main CRUD Operations

Clario will support the main CRUD operations required to manage business and client data.

Clients

- **Create** – Add a new client.
- **Read** – View the client list and full client record.
- **Update** – Edit client information.
- **Delete / Archive** – Remove or archive a client.

Appointments

- **Create** – Book a new appointment.
- **Read** – View appointments and available time slots.
- **Update** – Change appointment details or status.
- **Delete / Cancel** – Cancel an appointment.

Visits

- **Create** – Add a visit summary after an appointment.
- **Read** – View visit history and documentation.
- **Update** – Edit visit information.

Forms and Submissions

- **Create** – Create forms and submit client responses.
- **Read** – View forms and submissions.
- **Update** – Edit form definitions.
- **Delete** – Remove forms that are no longer in use.

Products

- **Create** – Add a product.
- **Read** – View products and recommendations.
- **Update** – Edit product details, price, or availability.
- **Delete / Deactivate** – Remove or deactivate a product.

Documents

- **Create** – Upload a document and create its database record.
- **Read** – View documents associated with a client or visit.
- **Delete** – Remove documents when permitted.

Purchases

- **Create** – Create a purchase and its purchase items.
- **Read** – View purchase history.
- **Update** – Update purchase status.

All operations will be protected by authentication and authorization rules.

5. API and Server Interface Design

Clario will primarily use Next.js Server Actions for communication between the frontend and server-side logic. Since the frontend and backend are part of the same Next.js application, a separate REST API is not required for most internal operations.

The main server operations will include:

- **Client Management** – createClient, getClients, getClientById, updateClient, archiveClient.
- **Appointment Management** – bookAppointment, getAppointments, getAvailableTimeSlots, updateAppointment, cancelAppointment.
- **Visit Management** – createVisit, getClientVisits, updateVisit.
- **Forms** – createForm, assignForm, submitForm, getForms, getFormSubmissions.
- **Documents** – uploadDocument, getClientDocuments, deleteDocument.
- **Products and Purchases** – createProduct, updateProduct, getProducts, getClientRecommendations, createPurchase, getPurchaseHistory.

Each server operation will validate input, verify the authenticated user's permissions, apply the required business logic, and read from or write to the database. API Routes / Route Handlers will only be added when a dedicated HTTP endpoint is required, mainly for communication with external services. For example, if online payments are implemented, a webhook endpoint may receive payment-status updates from the payment provider.

6. Core Business Logic

Clario's core business logic will enforce the rules required to keep the system consistent and ensure that users can only perform valid operations.

- **Appointment Booking** – Clients can only book available time slots within the business's defined availability. The server will recheck availability before creating an appointment to prevent double booking.
- **Business and Client Ownership** – Every client belongs to a specific business, and business owners can only manage data associated with their own business.
- **Visit Management** – A visit record is created for a relevant appointment and stores its professional documentation. Cancelled appointments will not create visit records.
- **Forms and Submissions** – Forms can only be assigned to clients belonging to the relevant business, and submissions are linked to both the client and form.
- **Document Access** – Documents are associated with a client and, when relevant, a specific visit. Clients can only access documents available to their own account.
- **Product Recommendations and Purchases** – Recommendations are linked to specific clients and may be associated with a visit. Purchases are linked to the

relevant client and business, and their total is calculated from the included products.

- **Reimbursement Preparation** – Clients can prepare reimbursement documentation only from visits and documents associated with their own client record.

These rules will be enforced on the server before data is created or modified.

7. State Management

Clario will use a simple state-management approach based on the type of data being handled.

- **Persistent Application Data** – Clients, appointments, visits, forms, products, and purchases will be stored in Supabase and loaded through the Next.js server layer.
- **Local UI State** – Temporary state such as open dialogs, selected tabs, filters, loading indicators, and selected dates will be managed within React components using `useState`.
- **Form State** – User input will be managed within the relevant form components. React Hook Form may be used if more complex form handling is needed.
- **Shared State** – Global state management will only be introduced if data later needs to be shared across many unrelated components.

8. Error Handling

Clario will handle errors at both the server and user-interface levels. The main error categories include validation errors, business-logic errors, authorization errors, database or storage failures, and errors from optional external services.

Server operations will return structured error results to the frontend. The interface will display clear user-friendly messages rather than technical details. For example, if an appointment becomes unavailable, the user will be asked to select another time.

Loading and failure states will be displayed when necessary, and unsuccessful operations will not be shown as successfully completed.

9. Input Validation

Clario will validate user input both in the interface for immediate feedback and on the server before data is stored.

- **Authentication and Client Information** – Required fields, valid email formats, and appropriate input lengths.

- **Appointments** – Valid dates and times, no appointments in the past, and compliance with business availability.
- **Forms** – Required questions and correct response types.
- **Products and Purchases** – Valid non-negative prices and quantities.
- **Documents** – Allowed file types and file-size limits.
- **Visit Information** – Required visit information before saving a record.

Zod may be used to define reusable validation schemas and maintain consistent validation across the application. Clear validation messages will be displayed next to the relevant fields.

10. Main User Experience Design - Clario will be designed as a simple and responsive web application with separate experiences for business owners and clients. The interface will prioritize clear navigation, quick access to common actions, and minimal unnecessary steps.

Business Owner Experience -The business-owner interface will focus on efficient daily management. The dashboard will provide quick access to upcoming appointments, clients, and common actions. Each Client Record will serve as a central workspace for accessing and managing all information related to a specific client.

Client Experience - The client interface will focus on simple self-service. Clients will be able to book and manage appointments, complete assigned forms, view visit information and documents, access product recommendations and previous purchases, and manage their personal information. Only relevant information and actions will be displayed.

General UX Principles

- Navigation and available actions will be adapted to the user's role.
- Frequently used actions will be easy to find and require as few steps as reasonably possible.
- Related information will be grouped and displayed consistently.
- Users will receive clear feedback after actions such as booking, saving, submitting, or cancelling.
- Validation and error messages will appear close to the relevant field or action.
- The interface will be responsive and usable on both desktop and mobile devices.

